ROHIT SONKER

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SUMMARY

Robotics Ph.D. candidate at CMU working on **decision-making agents, LLM based experimental design, reinforcement learning and multi-modal learning**. First-author work at **ICML, L4DC, and NeurIPS**, with algorithms experimentally validated in challenging scenarios such nuclear fusion tokamaks. Prior to the Ph.D., I have **4+ years shipping production LLM and ML systems** - RAG pipelines, agents, and cloud ML platforms to enterprise customers in life sciences and legal tech.

EDUCATION

Carnegie Mellon University — Ph.D. in Robotics, GPA 4.0/4.0 **Jan 2024 – Aug 2027**
Auton Lab · Advisor: Prof. Jeff Schneider · Pittsburgh, PA

Indian Institute of Technology (IIT) Kanpur — Dual Degree (B.Tech + M.Tech), Mechanical Engineering, GPA 9.3/10 **Jul 2014 – Jun 2019**

SELECTED PUBLICATIONS

[1] **R. Sonker**, X. Hu, H. Farre, A. Rothstein, J. Chen, E. Kolemen, J. Schneider. "Dynamics-Informed Adaptive Offline RL for Real-Time Tokamak Plasma Control." *NeurIPS 2026*

[2] EF Karagözlü *, **R Sonker***, T Gupta, B. Póczos, J. Schneider, "Bayesian Optimization with Rich Auxiliary Information via LLMs" *Arxiv 2026*

[3] Y. Fu, H. Bao, **R. Sonker**, X. Hu, A. Venugopal, J. Schneider, J.Chen, "Offline Reinforcement Learning for Plasma Control in Nuclear Fusion: Codebase and Benchmark", *NeurIPS 2026*

[4] **R. Sonker**, H. Farre, J. Chen, A. Rothstein, I. Char, R. Shousha, E. Kolemen, J. Schneider. "Offline Reinforcement Learning for Rotation Profile Control in Tokamaks." *Learning for Dynamics & Control (L4DC), 2026*

[5] T. Gupta, **R. Sonker**, E. M. Karagözlü, B. Póczos, J. Schneider. "Large Language Model-based Bayesian Optimization for Tokamak Stabilization." *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*

[6] **R. Sonker***, A. Capone*, A. Rothstein, H. Farre, E. Kolemen, J. Schneider. "Multi-Timescale Dynamics Model Bayesian Optimization for Plasma Stabilization in Tokamaks." *International Conference on Machine Learning (ICML), 2025*

[7] A. Rothstein, H. Farre, **R. Sonker**, S. K. Kim, A. Jalalvand, J. Schneider, E. Kolemen. "Preemptive Tearing Mode Suppression using Real-Time ECH Steering Machine Learning Stability Predictions on DIII-D." *Bulletin of the American Physical Society, 2024*

[8] V. Shaj, D. Büchler, **R. Sonker**, P. Becker, G. Neumann. "Hidden Parameter Recurrent State Space Models for Changing Dynamics Scenarios." *International Conference on Learning Representations (ICLR), 2022*

[9] **R. Sonker**, A. Dutta. "Adding Terrain Height to Improve Model Learning for Path Tracking on Uneven Terrain by a Four-Wheel Robot." *IEEE Robotics and Automation Letters (RA-L), 2021*

[10] **R. Sonker**, A. Mishra, P. Bansal, A. Pattnaik. "Techniques for Medical Concept Identification from Multi-Modal Images." *CEUR Workshop Proceedings, CLEF 2020 (Oral)*

EXPERIENCE

Graduate Research Assistant — Auton Lab, Carnegie Mellon University **Jan 2024 – Present**
Pittsburgh, PA

- **Multi-timescale Bayesian optimization** for sequential decision making under sparse data and expensive evaluations; deployed at the DIII-D National Fusion Facility, **improving instability avoidance by 117%** *(ICML 2025).*
- Built an **adaptive offline RL** trained using a learned **fusion world model** — latent-context-conditioned policies adapt to shifts in plasma dynamics; experimentally validated on the DIII-D tokamak *(L4DC 2026, NeurIPS 2026).*
- Designed an **LLM-guided Bayesian optimization** framework that elicits domain priors from LLMs and folds natural-language user feedback into parameter selection, improving convergence in scientific experimental design *(NeurIPS 2025 ML4PS).*
- Created a RL environment and benchmark for testing offline RL algorithm for Nuclear Fusion control tasks. *(NeurIPS 2026)*
- Built a **multimodal language–time-series model** with a Q-Former adapter and **LoRA** finetuning for cross-modal reasoning and question answering on small, domain-specific scientific datasets.
- Trained **DDPM diffusion models** for text-conditioned image generation, systematically comparing UNet and transformer backbones on scaling behavior, data efficiency, and sample quality.

Applied Sciences Intern — Microsoft, Office AI **Jun 2026 – Aug 2026**
Redmond, WA

- Developed an **Advantage Gated on-policy distillation algorithm** for selective distillation of agentic skills from multiple teachers, ensuring learning new skills while retaining performance of already learnt tasks.
- Created an RL-ready **gym environment for the Outlook agent** — containerized mailbox execution enabling parallelized rollouts across cluster nodes.

Tech Lead Manager, Machine Learning — CS DISCO **Aug 2023 – Jan 2024**

Legal technology · Gurgaon, India

- Led the India data science team (3 direct reports) designing **LLM agents** for conversational search, summarization, and document tagging over legal discovery data — shipped the features that landed **DISCO's first paid AI customers**.
- Upgraded a **RAG retrieval chatbot** with conversational memory and pronoun coreference resolution for accurate long-context, multi-turn retrieval, yielding an **18% gain in accuracy and task completion**.

Senior Data Scientist — PricewaterhouseCoopers (PwC) US Advisory **Jul 2019 – Jul 2023**

Pharma & Life Sciences · Mumbai, India

- Led development of an **LLM system generating regulatory-compliant clinical trial documents** from tabular trial parameters; benchmarked GPT-3.5 prompting against fine-tuned T5, BERT, and GPT models and deployed into client infrastructure — **secured $2M+ in engagements**.
- **Architected the healthcare practice's AWS ML platform** — serverless ETL (Spark, S3, RDS), SageMaker development environment, EC2 deployment, third-party integrations — unifying 10+ projects and data sources and training 50+ team members.
- Built a **clinical trial design system** combining predictive models, simulation, and LLM agents; adopted across multiple large pharma clients for **$0.5M+ in revenue**.
- Productionized an unsupervised **NLP entity-matching pipeline** end to end on AWS (Step Functions, SageMaker, CI/CD, SNS) supporting both scheduled batch jobs and on-demand API inference, plus a **model-drift monitoring framework** for deployed models.

EARLIER RESEARCH

Visiting Researcher — Karlsruhe Institute of Technology **Jul 2021 – Dec 2021**

Advisor: Prof. Gerhard Neumann · Remote

- Developed **latent-context recurrent state-space models** that infer unobserved context on the fly in changing-dynamics settings, outperforming meta-learning baselines (ICLR 2022).
- Validated the models on control tasks by learning inverse dynamics for a Franka Emika Panda manipulator and a pneumatically actuated muscle robot.

Master's Thesis — Indian Institute of Technology Kanpur **Jul 2018 – Jun 2019**

Advisor: Prof. Ashish Dutta · Kanpur, India

- Combined **model learning with model predictive control** for path tracking by a four-wheel mobile robot, introducing terrain-height context to generalize across unseen terrains (IEEE RA-L 2021).
- Ran ablations over planning horizon, controller frequency, and cross-entropy-method action sampling to isolate the sources of tracking improvement.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, R, MATLAB, Spark

ML & AI: PyTorch, scikit-learn, NumPy; offline & model-based RL, Bayesian optimization / Gaussian processes, diffusion models, LLM post-training (SFT, RLHF, distillation, LoRA/PEFT), multimodal models, RAG, LangChain, FAISS / Pinecone

ML Infra: AWS (SageMaker, Step Functions, Lambda, S3, RDS, EC2), Azure, Docker, Kubernetes, MLflow, DVC, CI/CD

Simulation: ROS, MuJoCo, PyBullet

Analytics: Pandas, Tableau, PowerBI, Streamlit, RShiny

TALKS & HONORS

- **Oral talk** — "Dynamics Model Bayesian Optimization for Experimental Design in Tokamaks," APS Division of Plasma Physics (APS-DPP), 2025.
- **2nd place worldwide** — ImageCLEF 2020 medical concept detection challenge (multi-label prediction over radiographic images); results presented at CLEF 2020.